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THE INFLUENCE OF NEGATIVE VALUES OF VARIABLES IN THE CALCULATION OF VSL TO FIND A WAY OUT OF THE ECONOMIC CRISIS

Pil Eduard Anatolyevich

Academician RANH, Doctor of Technical Sciences, Full Professor

Saint-Petersburg State University of Aerospace Instrumentation, Russia

Annotation. *The article deals with the issue of the way out of any country's economic crisis, which periodically arises and covers almost all countries of the world in one way or another. The author offers his vision of this issue on the basis of the theory of economic crises developed by him. The presented material suggests ways out of the economic crisis using negative values of the variables under consideration in relation to the spherical economic shell.*

Keywords: *gross domestic product, recovery from the economic crisis, calculations in Excel, 2D and 3D drawings.*

Earlier in his works, the author showed that the GDP of any country can be represented as the volume of a spherical or ellipsoid shell [1, 2, 3]. In these books, it was shown that when exposed to external pressure on these shells, with a number of values of the variables under consideration, their volume can not only shrink, but even increase or remain constant. Thus, by selecting the appropriate values of variables, we can quickly get out of the economic crisis with the growth of economic indicators.

The question of using negative values of variables in an economic crisis is discussed below. The author has previously published a number of articles on this issue [4, 5, 6, 7, 8]. And it would seem that this is an absurd question, but it takes place, for example, the banks of Sweden, Switzerland, Denmark and Japan issued such loans to some customers [9, 10, 11].

It should be remembered here that in many countries, central banks, when the next global economic crisis came, issued money to banks, enterprises and even the population in order to stimulate the country's economy in this way. Last year, the Nobel Prize in Economics was awarded to three American economists who have significantly improved the understanding of the role of banks in the economy, especially during financial crises.

This article considers the issue of calculating the values of the lower critical volume of the Vsl spherical economic shell and constructing 2D and 3D drawings when the values of three variables change, i.e. $Vsl (GDPsl) = f(X1sl, X2sl, X3sl)$. In some cases, when the Vsl values have small minima and maxima compared to the last calculated Vsl value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

So, Figure 1 and 2 show two Vsl figures, which were constructed with the following values of variables $X1sl = 1$, $X2sl = 1...10$, $X3sl = 5...-5$. As can be seen from the figures presented, the Vsl values grow from 0.02 to a maximum value of 76.98, after which they fall to zero, because with negative values of the variable $X3sl$ they have no solutions. In this example, the value of the Vsl parameter is increased by 3334.9 times.

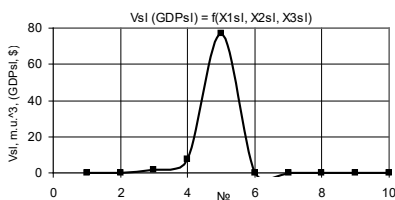


Figure 1. 2D for Vsl (GDPsl)

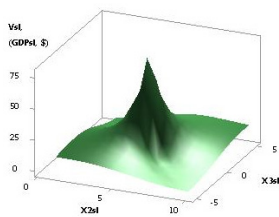


Figure 2. 3D for Vsl (GDPsl)

The following two figures 3 and 4 were constructed with variables $X1sl = 1$, $X2sl = 1...10$, $X3sl = 5...-5$. As can be seen from Fig. 3, here the Vsl values increased significantly from 0.02 to 860.63, i.e. 37285.33 times.

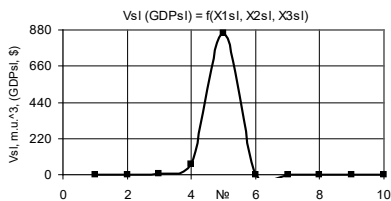


Figure 3. 2D for Vsl (GDPsl)

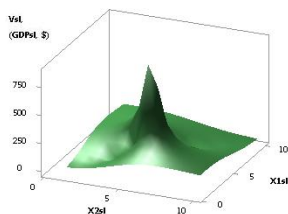


Figure 4. 3D for Vsl (GDPsl)

If we use the following values of the variables $X1sl = X2sl = 1...10$, $X3sl = 5...-5$, then in this case the maximum value of Vsl = 6.89, as can be seen from Fig. 5 and at the same time the value increased by 298.28 times.

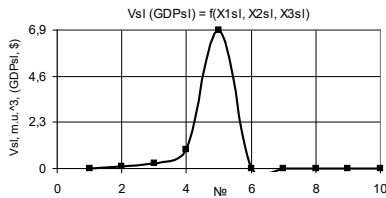


Figure 5. 2D for Vsl (GDPsl)

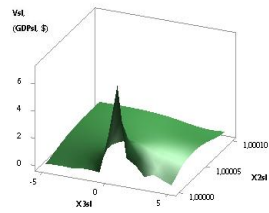


Figure 6. 3D for Vsl (GDPsl)

The following two figures 7 and 8 were constructed with variables $X1sl = 1$, $X2sl = 5 \dots -5$, $X3sl = 1 \dots 10$. As can be seen from Figure 7, here the Vsl values have a minimum of 0.003 at point 6.

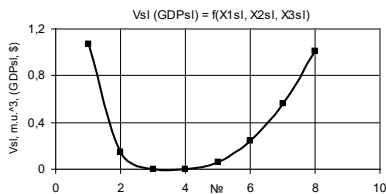


Figure 7. 2D for Vsl (GDPsl)

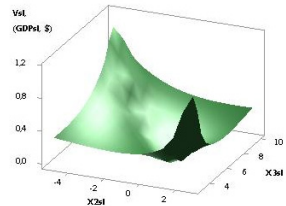


Figure 8. 3D for Vsl (GDPsl)

If the following values of the variables $X1sl = X3sl = 1 \dots 10$, $X5sl = 5 \dots -5$ are applied in the calculations, then in this case the Vsl parameter will also have a minimum of 0.04 at point 6, as can be seen from Fig. 9. At the same time, the constructed graph turned out to be symmetrical.

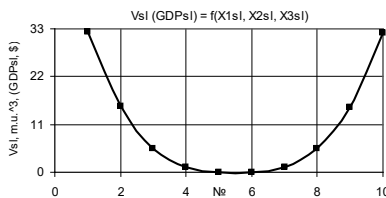


Figure 9. 2D for Vsl (GDPsl)

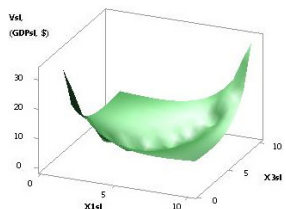


Figure 10. 3D for Vsl (GDPsl)

When constructing the following two figures 11 and 12, the variables $X1sl = 5 \dots -5$, $X2sl = 1 \dots 10$, $X3sl = 1$ were used. As can be seen from Figure 11, here the Vsl values have a maximum of 35.64 at point 4.

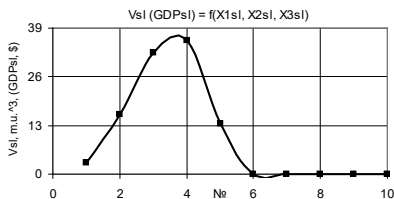


Figure 11. 2D for Vsl (GDPsl)

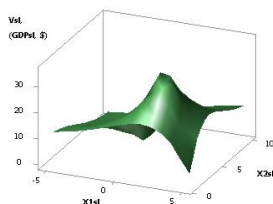


Figure 12. 3D for Vsl (GDPsl)

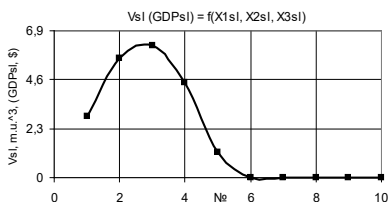


Figure 13. 2D for Vsl (GDPsl)

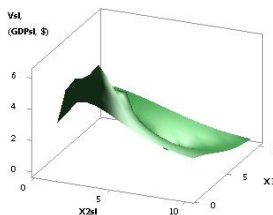


Figure 14. 3D for Vsl (GDPsl)

The following two figures 13 and 14 were constructed when the variables were $X1sl = 5 \dots -5$, $X2sl = X3sl = 1.10$. As can be seen from Figure 13, here the Vsl values also have a maximum of 6.22 at point 3.

The following Figures 15 and 16 clearly show the changes in the calculated Vsl parameter when the variables had values $X1sl = 1$, $X2sl = 10 \dots 1$, $X3sl = 5 \dots -5$. In this example, the constructed curve of the Vsl parameter increases to a maximum value of 133.02 at point 5, after which its values fall to zero at point 6.

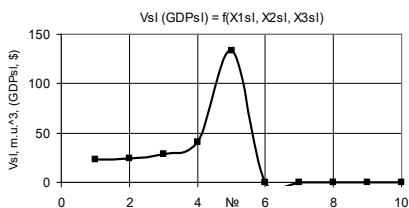


Figure 15. 2D for Vsl (GDPsl)

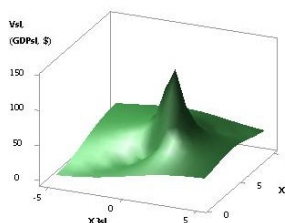


Figure 16. 3D for Vsl (GDPsl)

If the values of the variables are as follows $X1sl = X2sl = 10 \dots 1$, $X3sl = 5 \dots -5$, then in this example, the built Vsl dependency in Figure 17 has a minimum of 642.02 at point 3, after which it increases to a maximum value of 1954.94 at point 5. At point 6, the Vsl parameter drops to zero.

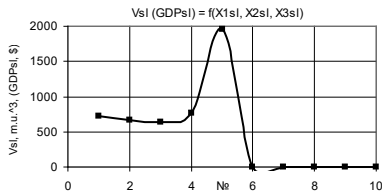


Figure 17. 2D for Vsl (GDPsl)

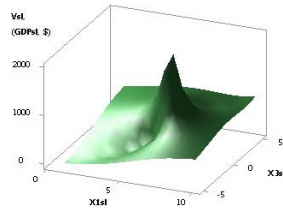


Figure 18. 3D for Vsl (GDPsl)

In the following example, the values of the variables $X1sl = 10...1$, $X2sl = 1...10$, $X3sl = 5...-5$ were used to construct the Vsl curve. Here in Figure 19, the Vsl parameter grows between points 1-5 from 0.73 to 1131.33, i.e. 1549.32 times, after which it drops to zero at point 6.

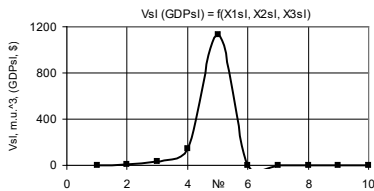


Figure 20. 2D for Vsl (GDPsl)

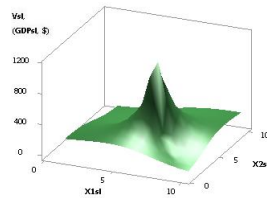


Figure 21. 3D for Vsl (GDPsl)

If, when constructing two figures 21 and 22, we use the values of the variables $X1sl = 10...1$, $X2sl = 5...-5$, $X3sl = 1...10$, then in this case we get the Vsl dependence, which will have a minimum of 0.03 at point 3 and a maximum of 1.58 at point 6.

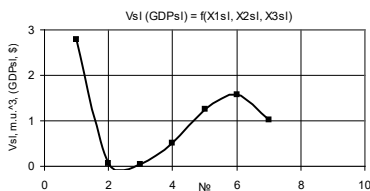


Figure 21. 2D for Vsl (GDPsl)

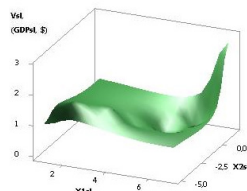


Figure 22. 3D for Vsl (GDPsl)

In the following two figures 23 and 24, dependencies were constructed for $X1sl = 1.10$, $X2sl = 5...-5$, $X3sl = 10.1$. Here, too, the Vsl parameter values have a maximum of 1.59 at point 2 and a minimum of 0.03 at point 5.

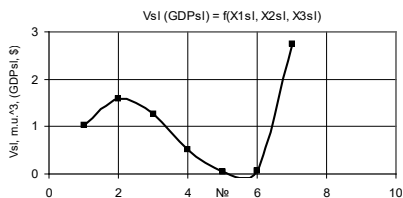


Figure 23. 2D for Vsl (GDPsl)

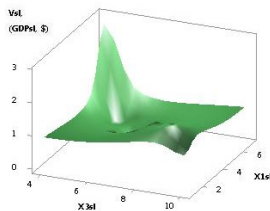


Figure 24. 3D for Vsl (GDPsl)

If we use the following values of the variables $X1sl = 1...10$, $X2sl = 5...-5$, $X3sl = 1...0.1$, then in this case the Vsl curve in Figure 25 has a minimum of 1.36 at point 6 and the graph also turned out to be symmetrical.

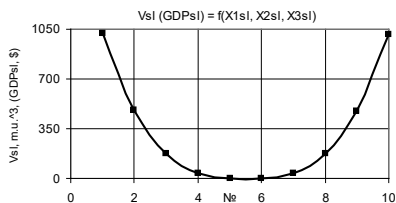


Figure 25. 2D for Vsl (GDPsl)

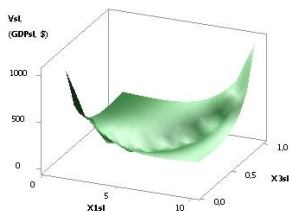


Figure 26. 3D for Vsl (GDPsl)

The following two figures 27 and 28 were constructed when the variables were $X1sl = 1...0.1$, $X2sl = 5...-5$, $X3sl = 1...10$. Here in Figure 27, the Vsl curve presented has a minimum of 0.001 at point 3 and a maximum of 0.05 at point 6.

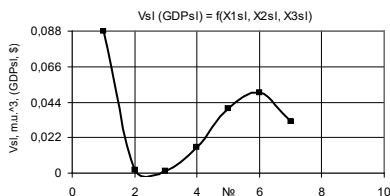


Figure 27. 2D for Vsl (GDPsl)

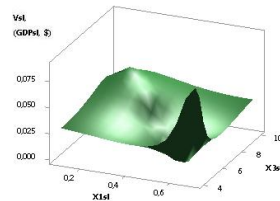


Figure 28. 3D for Vsl (GDPsl)

With variables $X1sl = 1...10$, $X2sl = 5...-5$, $X3sl = 1...0.1$, the constructed Vsl curve in Figure 29 also has both a maximum of 50.32 at point 2 and a minimum of 1.09 at point 5.

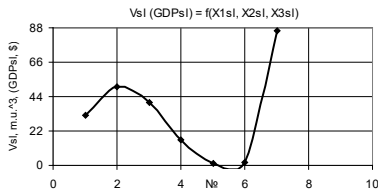


Figure 29. 2D for Vsl (GDPsl)

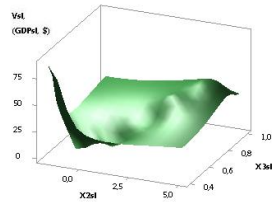


Figure 30. 3D for Vsl (GDPsl)

The last two Figures 31 and 32 also show the dependencies Vsl (GPSsl) = $f(X1sl, X2sl, X3sl)$, which were constructed with the following values of the variables $X1sl = 1 \dots 0.1$, $X2sl = 5 \dots -5$, $X3sl = 10.1$. Here the Vsl curve first grows from a value of 0.032 to a maximum 0.05, and then falls to a minimum of 0.001 at point 5, decreasing by 46.16 times.

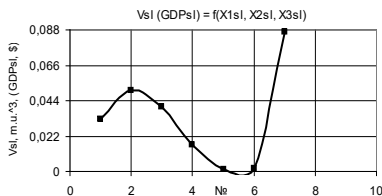


Figure 31. 2D for Vsl (GDPsl)

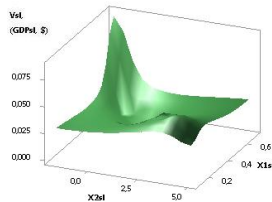


Figure 32. 3D for Vsl (GDPsl)

Thus, as can be seen from the Figures presented above, the values of the Vsl parameter, despite the fact that negative variables were used, can both rise and fall. Therefore, by choosing specific values of variables and financing them, the state, when the next economic crisis is just beginning to develop, can prevent it in the country at the very beginning. At the same time, much less money will be spent than to do this already in the middle of the economic crisis. It should also be mentioned here that in some cases, variable values of variables in no way change the calculated values of the volume of the Vsl economic shell, the values of which remain constant and equal to 0.26 ($Vsl = 0.26$). This can be clearly seen from tables 1 and 2 below. That is, these values of variables should not be used when considering options for overcoming the economic crisis.

Table 1. Calculation parameter Vsl
 at $X1sl = X3sl = 5 \dots -5$, $X2sl = -1$

№	X1sl, m.u.	X2sl, m.u.	X3sl, m.u.	Vsl, m.u. ³ (GDPsl, \$)
1	5,0	1	5,0	0,26
2	3,9	1	3,9	0,26
3	2,8	1	2,8	0,26
4	1,7	1	1,7	0,26
5	0,6	1	0,6	0,26
6	-0,6	1	-0,6	0,26
7	-1,7	1	-1,7	0,26
8	-2,8	1	-2,8	0,26
9	-3,9	1	-3,9	0,26
10	-5,0	1	-5,0	0,26

Table2. Calculation parameter Vsl
 at $X1sl = X3sl = 5 \dots -5$, $X2sl = -1$

№	X1sl, m.u.	X2sl, m.u.	X3sl, m.u.	Vsl, m.u. ³ (GDPsl, \$)
1	5,0	-1	5,0	0,26
2	3,9	-1	3,9	0,26
3	2,8	-1	2,8	0,26
4	1,7	-1	1,7	0,26
5	0,6	-1	0,6	0,26
6	-0,6	-1	-0,6	0,26
7	-1,7	-1	-1,7	0,26
8	-2,8	-1	-2,8	0,26
9	-3,9	-1	-3,9	0,26
10	-5,0	-1	-5,0	0,26

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